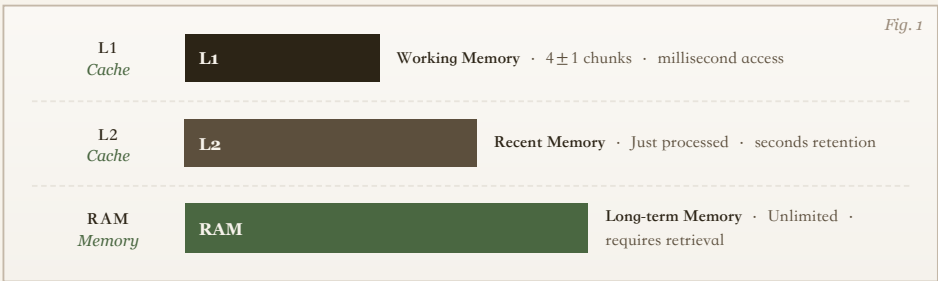


Working Memory is Like CPU Cache

A Minimal Model for Understanding How the Brain Learns · Printable Plate

Analogia: Architectura Memoriae Hierarchica

I. THE ANALOGY



A CPU does not read directly from the hard drive; it relies on hierarchical caches. Your brain works the same way: what you use for thinking is not long-term memory, but extremely limited working memory. When the cache overflows, a "cache miss" occurs — you feel confused, distracted, unable to comprehend. This is not stupidity; it is an architectural constraint.

II. OBSERVATIONS

Obs. 1

Chunking

Compress scattered information into "chunks," like zipping multiple files. The gap between experts and novices is essentially the size of their chunks.

Obs. 2

Spaced Repetition

Do not cram all at once; let the cache "warm up." Spaced retrieval is $2-3\times$ more efficient than continuous reading.

Obs. 3

Single Tasking

Context-switching is extremely costly — each switch clears the cache. Focus is performance optimization.

Obs. 4

Building Connections

New knowledge must "mount" onto existing cache. Isolated information is immediately discarded.

METHODS

- 1** One instruction at a time — for children, colleagues, and yourself. Break down multi-step tasks.
- 2** Framework first, details second — an outline is a signpost in the L1 cache, preventing information overwhelm.
- 3** Explain the new with familiar analogies — "It's like..." is not laziness; it helps the brain build cache mappings.
- 4** Allow mind-wandering — cache cleanup is necessary. Forcing continuous focus lowers hit rates.

